



ISTANBUL TECHNICAL UNIVERSITY

FACULTY OF COMPUTER AND INFORMATICS

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA ENGINEERING**

AI AND DATA ENGINEERING DESIGN

PROJECT CHARTER I

DESCRIPTION

This form must be filled by considering only the first part of the design project.

The first part of the AI and Data Engineering Design Course includes the following tasks:

- Determining a complex engineering design problem with multiple constraints.
- Conducting a comprehensive literature survey.
- Setting and analyzing design constraints.
- Determining and analyzing relevant engineering standards.
- Analyzing alternative solutions.
- Assessing relevant risks.
- Developing a requirement analysis report
- Developing the project work plan and schedule for completion of the project.

The project must focus on AI and Data Engineering and should be related to multiple disciplines or have a significant impact in various fields. Some examples:

Computer-Aided Paper Bill Classification for Visually Impaired Persons.

Social Media Monitoring and Analysis for Stock Market Trading.

Generative Language Modeling for Automated Theorem Proving.

Forecasting Extreme Weather Events from Local and Global Weather Data.

Generative Models for Fashion Products.

Automated Detection and Classification of Pneumonia in X-Rays.

1. PROJECT TEAM

ID	NAME	MAJOR
150220312	Alper Düzgün	AI and Data Engineering
150220754	Hazar Utku Sözer	AI and Data Engineering

2. PROJECT NAME

Advanced Model Compression Techniques for Resource-Constrained AI Architectures

3. SHORT SUMMARY (ABSTRACT)

This project focuses on the research, analysis, and implementation of advanced model compression techniques to reduce the computational and memory footprint of modern deep learning architectures. The primary aim is to systematically evaluate methods such as weight sharing (e.g., HalViT), quantization, and pruning to optimize models for deployment in resource-constrained environments. This multi-disciplinary problem requires balancing theoretical linear algebra with software engineering and hardware optimization. The design process will be governed by strict constraints, including preserving baseline accuracy, ensuring framework interoperability, and achieving measurable reductions in computational overhead. Measurable objectives include benchmarking compressed models against uncompressed baselines regarding total parameter count, GFLOPs, and task-specific accuracy metrics. The evaluation framework will align with engineering standards such as ITU-T F748.11 for neural network processing evaluation and ISO/IEC 22989:2022 for AI terminology.

The primary aim is to reduce the memory usage and computational complexity of a deep learning model without experiencing severe losses in model accuracy by using model compression techniques.

4. KEYWORDS

Model Compression

Pruning

Quantization

Knowledge Distillation

Parameter Efficiency

Weight Sharing

Edge AI

5. DESIGN CONSTRAINTS

Efficiency/Resource Constraint: The engineered solution must demonstrate a mathematically verifiable reduction in either total parameter count, memory footprint (MB/GB), or computational complexity (GFLOPs) to qualify as a successful compression implementation.

Interoperability Constraint: The architectural modifications and resulting compressed models must remain compatible with standard, open-source deep learning frameworks (e.g., PyTorch) to ensure the research is reproducible and measurable by standard academic tools.

Functionality/Accuracy Constraint: The application of compression techniques must preserve the core utility of the baseline model. The top-1 accuracy (or perplexity, if evaluating language models) must not degrade beyond a predefined, acceptable threshold (e.g., < 2% drop) compared to the uncompressed baseline.

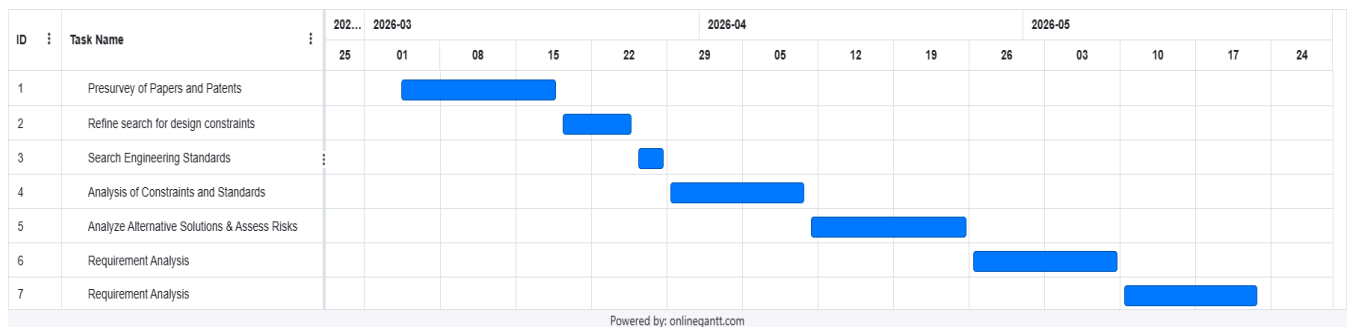
6. RELEVANT ENGINEERING STANDARDS

ITU-T F748.11 (08/2020): Metrics and evaluation methods for a deep neural network processor benchmark. This will guide how the project officially measures memory footprint and computational efficiency.

ISO/IEC 22989:2022: Information technology — Artificial intelligence concepts and terminology. This standard ensures the terminology used in the requirement analysis and final reporting aligns with global industry specifications.

7. WORK BREAKDOWN, DELIVERABLES AND TIME PLAN

#	TASK	START DATE	DUE DATE	DELIVERABLE
1	Presurvey of Papers and Patents	04-03-2026	18-03-2026	Literature Survey
2	Refine search for design constraints	19-03-2026	25-03-2026	Analysis of Design Constraints in Literature
3	Search Engineering Standards	26-03-2026	28-03-2026	Listing of Relevant Engineering Standards
3.1	Analysis of Constraints and Standards	29-03-2026	10-04-2026	
3.2	Analyze Alternative Solutions & Assess Risks	11-04-2026	25-04-2026	
4	Requirement Analysis	26-04-2026	09-05-2026	
5	Requirement Analysis	10-05-2026	22-05-2026	Report Draft



8. INTELLECTUAL PROPERTY RIGHTS

Important: The copyright ownership of reports and other project documents produced during the design course project belongs to the student, the project advisor(s), the AI and Data Engineering Department, and Istanbul Technical University. Any additional materials, such as scientific publications, patents, inventions, or commercial products, arising from the project are subject to Istanbul Technical University's Intellectual Property Regulations and the laws of the Turkish Republic. If a third institution or company is involved in the design project, the ownership of inventions and intellectual property rights must be determined

through a special signed agreement between the department and the parties involved.

9. SUPERVISOR INFO AND CONFIRMATION

Date of Proposal	02.02.2026
Academic Term of Project Delivery	2026 Spring Term
Student(s) Signature	Alper Düzgün Hızır İhsan ÖZEL Alper Hızır
Project Advisor(s) Name	Prof. Dr. Behçet Uğur Töreyn
Project Advisor(s) Signature	